Computer Science & IT



Database Management System

Transactions & concurrency control

Lecture No. 11



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Recap of Previous Lecture



- **Topic** Concurrency control protocols
- **Topic** Simple use of shared and exclusive locks
- **Topic** Two phase lock (2PL)

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Topics to be Covered



- Topic: Basic 2PL and lock upgrading/downgrading
- Topic: Problems possible with Basic 2PL
- Topic: Strict 2PL
- Topic: Conservative 2PL
- Topic: Rigorous 2PL

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Topic : Basic 2PL with lock upgrading

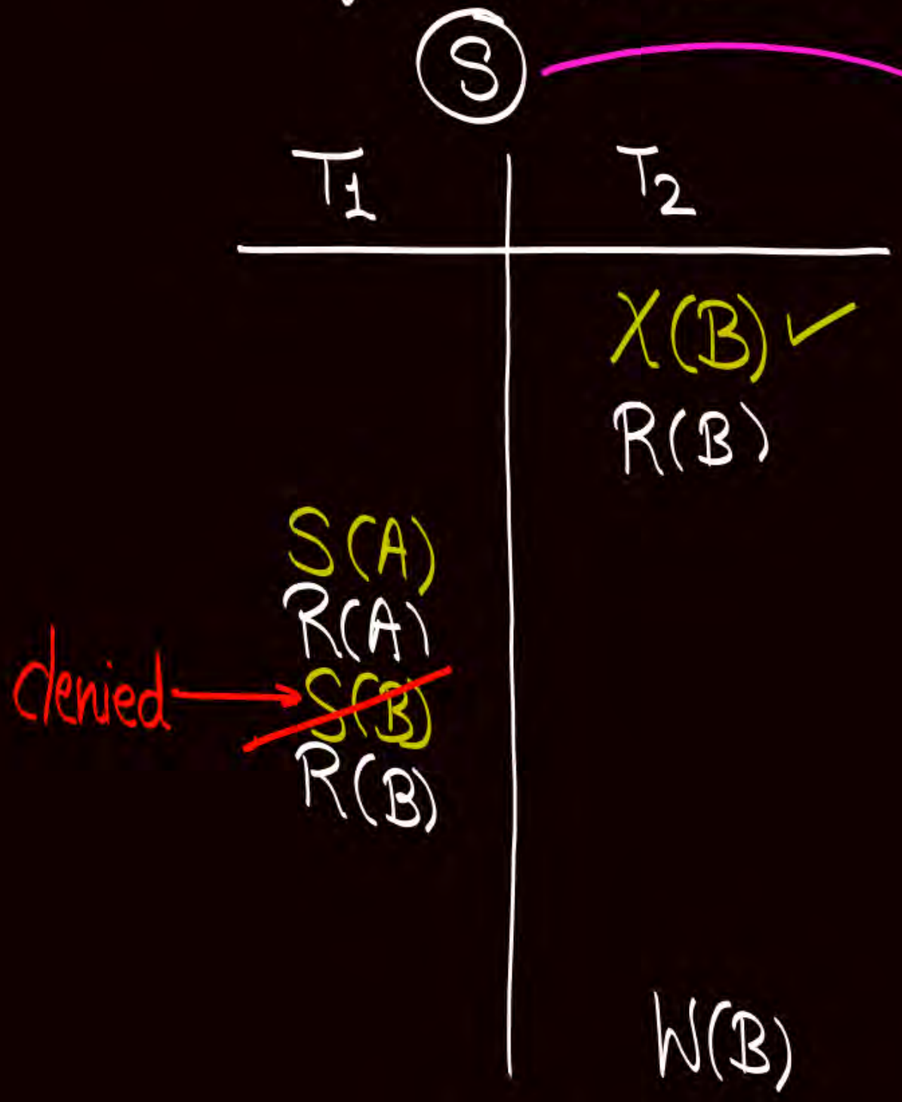
If lock conversion (upgrading/downgrading) is allowed, then we can upgrade shared lock into exclusive locks in the Growing Phase, and we can downgrade from the exclusive lock to shared lock in the shrinking phase.

Upgrade ⇒ Shared to Exclusive

Downgrade ⇒ Exclusive to Shared

* We are not allowed to upgrade if we are already in the shrinking phase
* If transaction is not already in its shrinking phase, then also we can downgrade the lock, but as soon as we downgrade shrinking phase will start.

Q:- Check whether the following schedule is allowed to execute using Basic 2PL or not ?



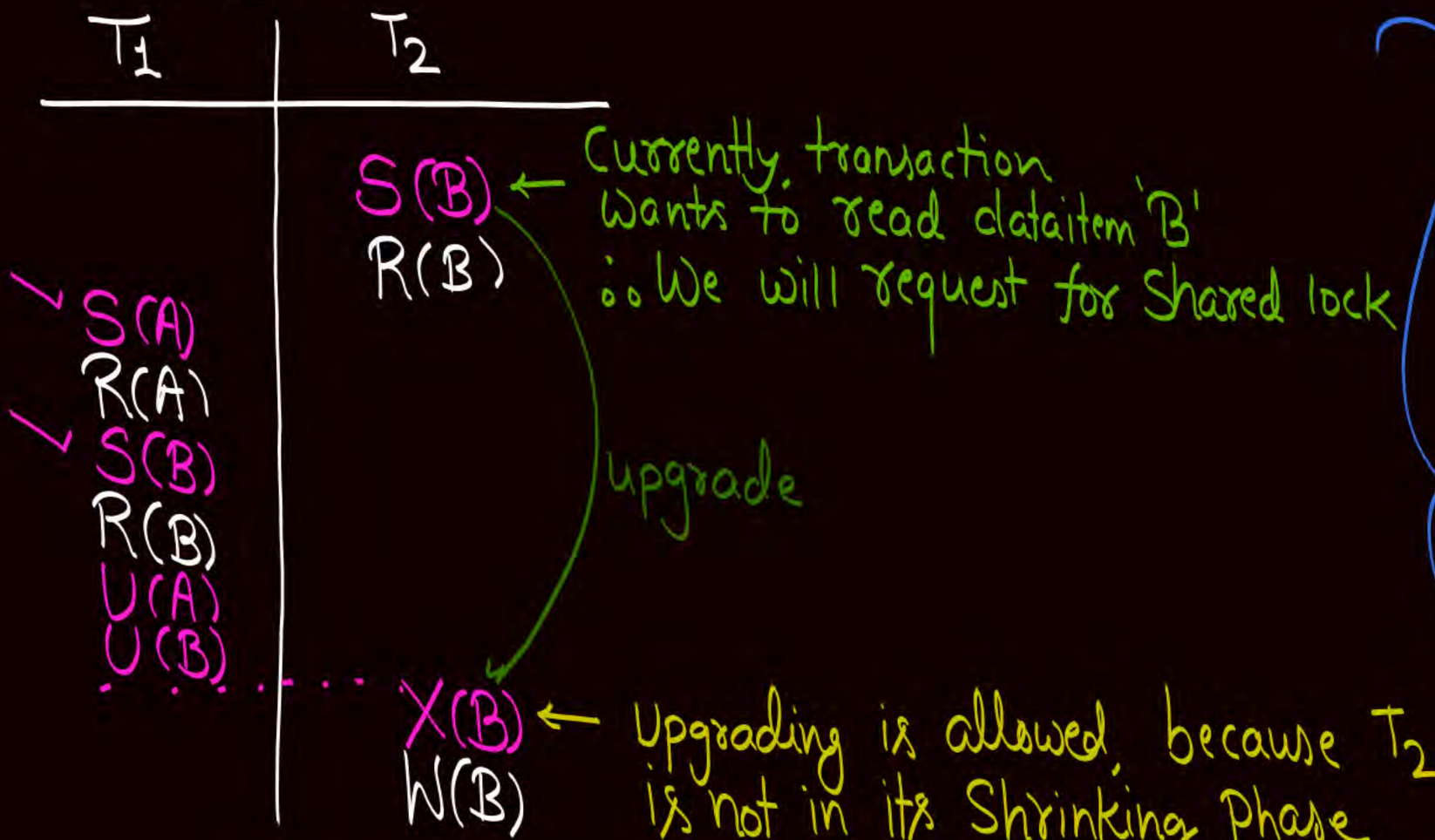
It is a Conflict Serializable Schedule

↓

and it is not allowed to execute using Basic 2PL (without lock upgrade)

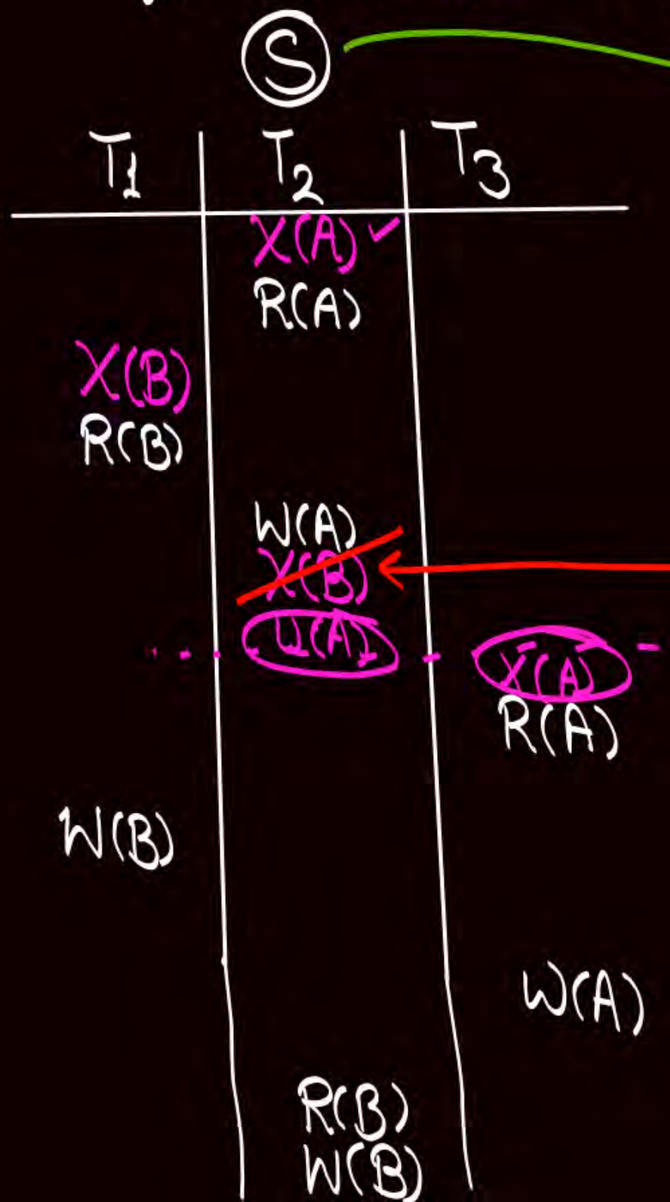
Q:- Check whether the following schedule is allowed to execute using Basic 2PL with lock upgrading or not?

(S)



It is a C.S.S. which is not allowed by Basic 2PL without lock upgrade but it is allowed by Basic 2PL with lock upgrade

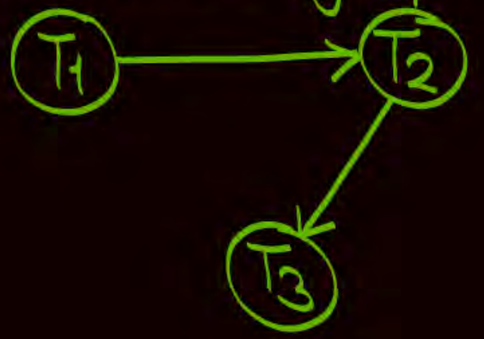
Q:- Check whether the following schedule is allowed to execute using Basic 2PL or not ?



denied

∴ Not allowed by Basic 2PL without lock upgrade

Precedence graph



⇒ Acyclic

∴ C.S.S.

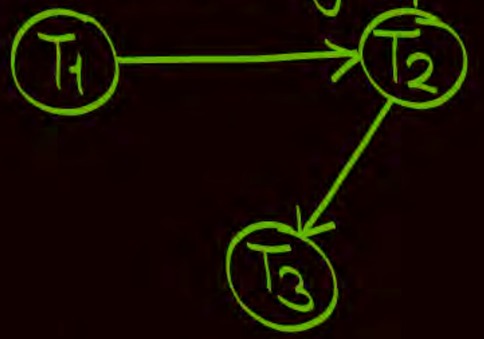
but not allowed by Basic 2PL without lock upgrading

Q:- Check whether the following schedule is allowed to execute using Basic 2PL with lock upgrading or not?



denied → ∴ Not allowed by Basic 2PL even when lock upgrading is allowed

Precedence graph



⇒ Acyclic

∴ C.S.S.

but not allowed by Basic 2PL even when lock upgrading is allowed

Note:- ① If lock upgrading is allowed, then we may execute some extra conflict serializable schedules Compared to Basic 2PL without lock upgrading, but still there remains some conflict serializable schedules that are not allowed by Basic 2PL even when lock upgrading is allowed

② If schedule is not a conflict serializable schedule then it is never allowed by any type of 2PL, does not matter whether lock upgrading is allowed or not.



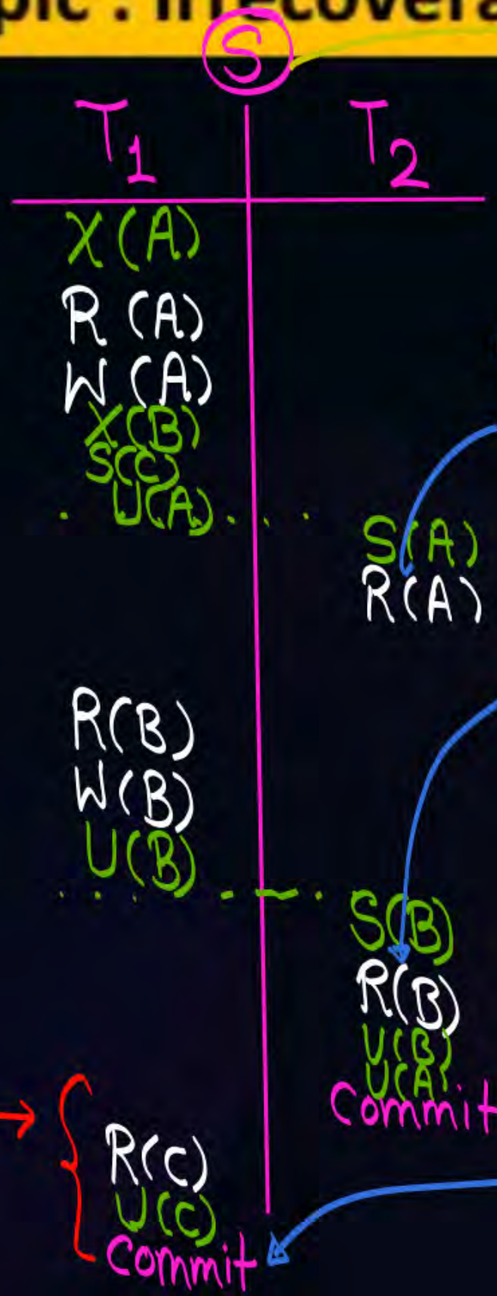
Topic : Problems possible with Basic 2PL

A schedule that is allowed to execute using basic 2PL protocol may suffer from,

1. Irrecoverability { Can be solved using Strict-2PL }
2. Deadlock { Can be solved using Conservative 2PL }
3. Starvation { No solution }

Topic : Irrecoverability with Basic 2PL





It is a conflict serializable schedule which is allowed by Basic 2PL. But it is an irrecoverable schedule

Reads the value written by Uncommitted transaction T_1

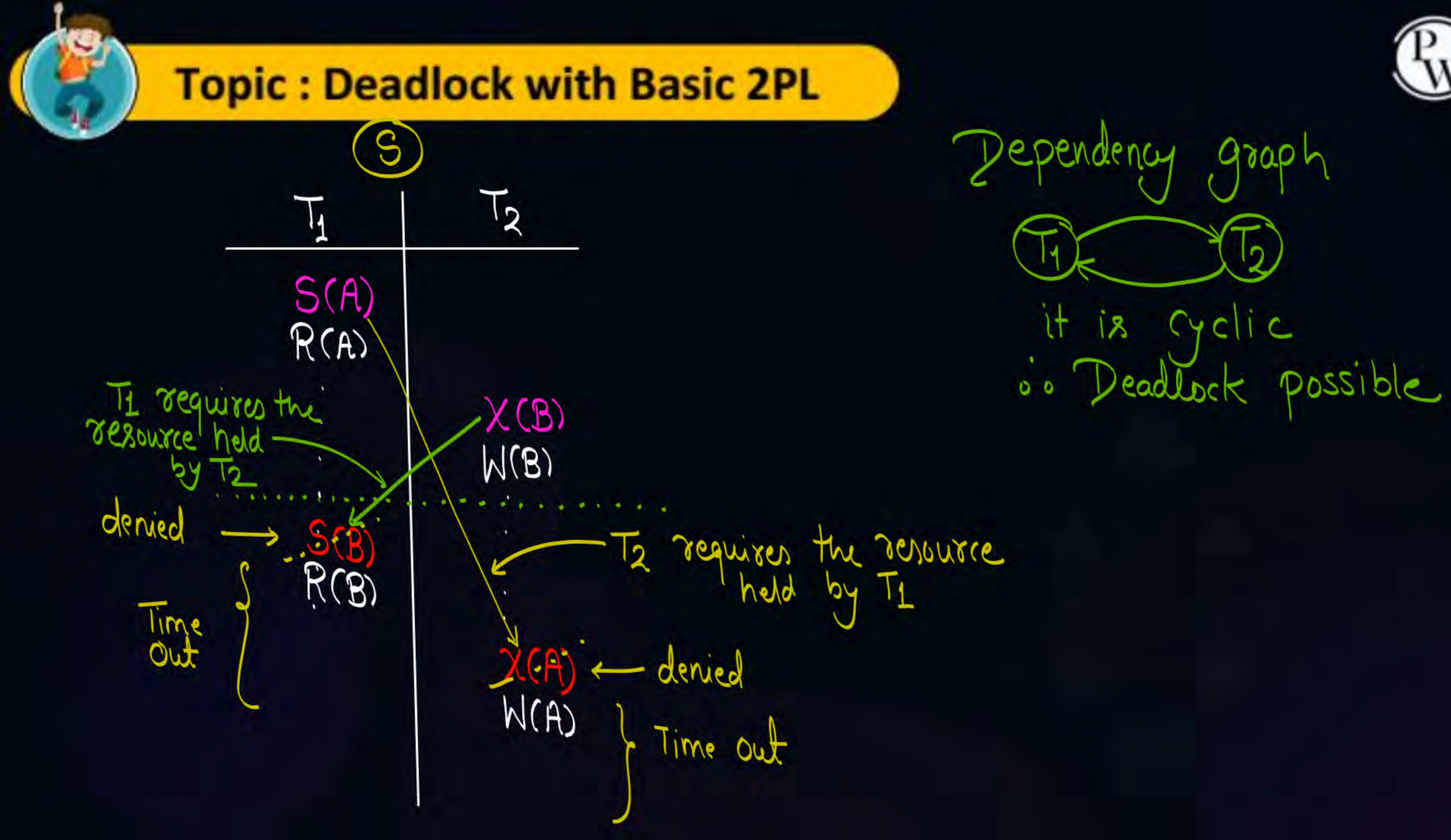
$\therefore$ T_2 depends on T_1

Unsafe →

Dependent transaction (i.e. T_2) Commit before the Commit/Rollback of transaction on which it depends {i.e. T_1}

$\therefore$ Irrecoverable Schedule

Topic : Deadlock with Basic 2PL

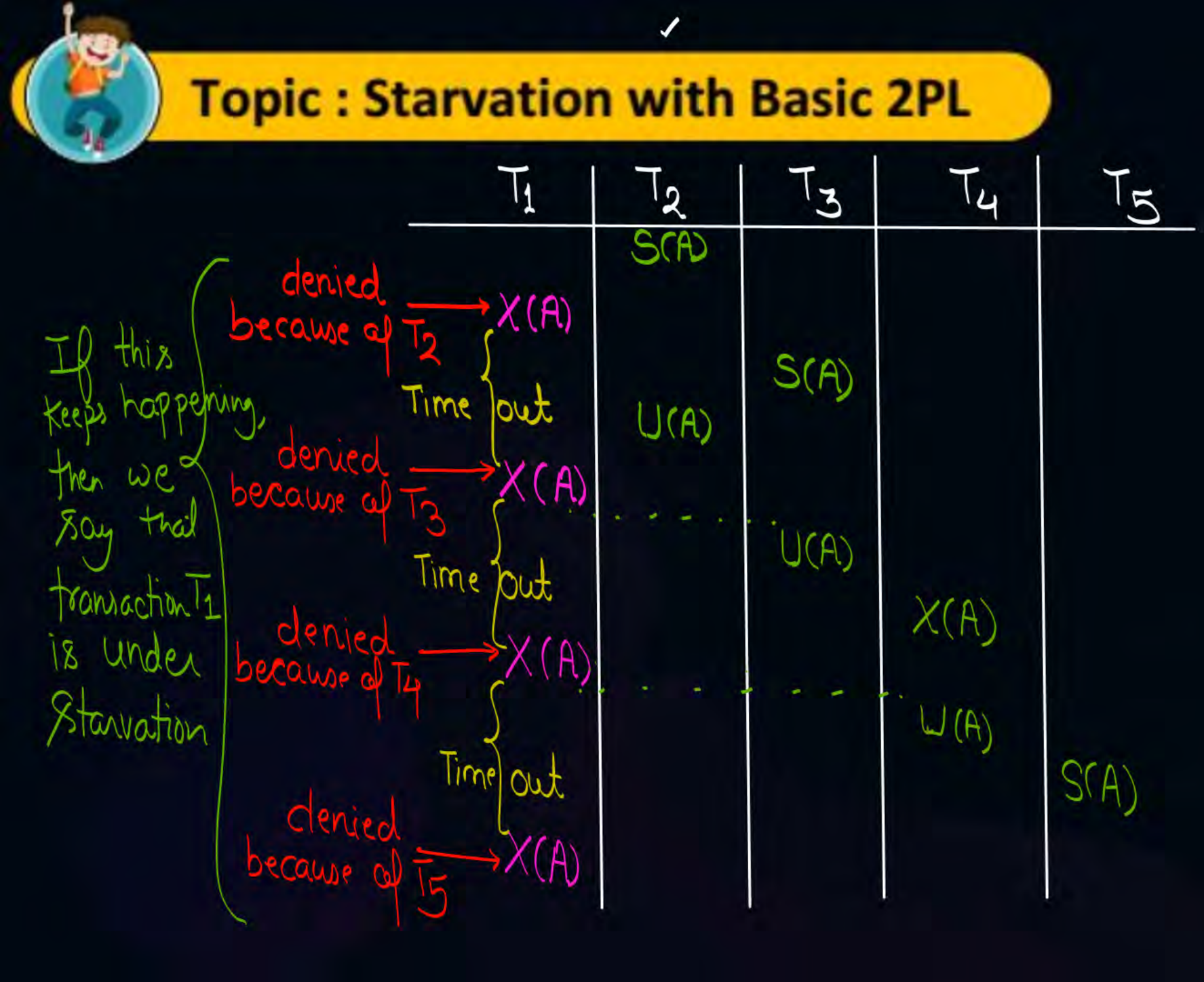


Dependency graph

it is cyclic

∴ Deadlock possible

Topic : Starvation with Basic 2PL





2 mins Summary

- **Topic** Basic 2PL and lock upgrading/downgrading
- **Topic** Problems possible with Basic 2PL
- **Topic** Strict 2PL
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- **Topic** Rigorous 2PL

THANK - YOU

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